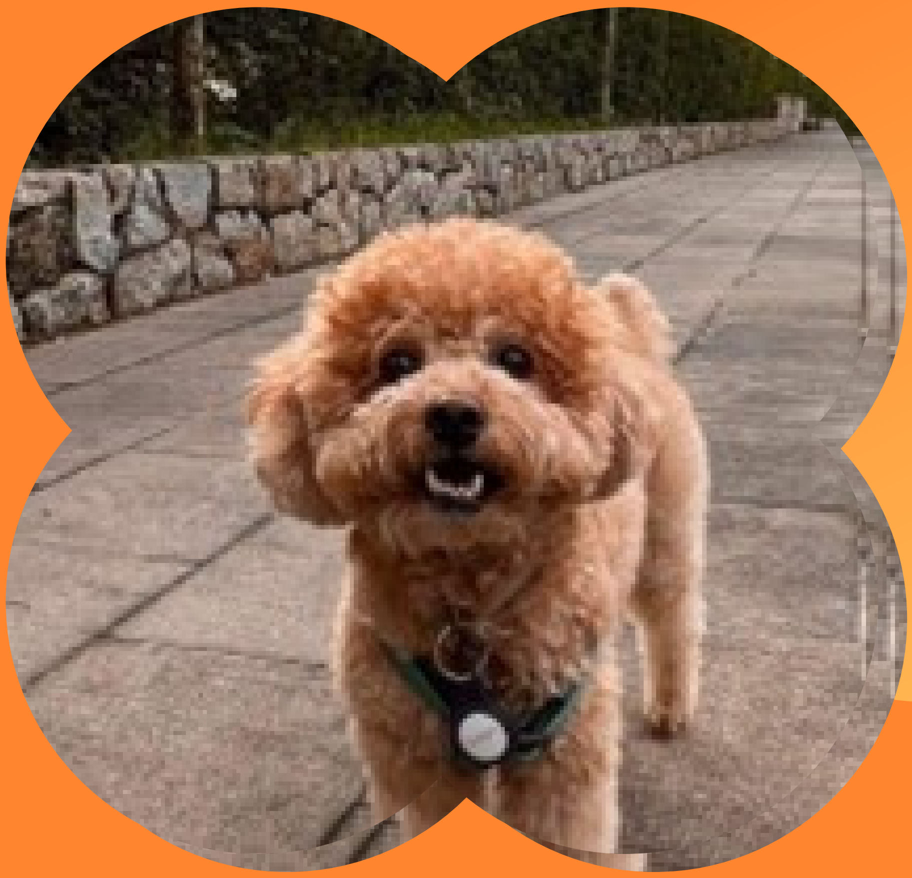
DOUGLAS COLLEGE

WINTER 2025

Project Proposal

"FurBot"

A web-based chatbot offering quick and accurate assistance for dog owners' everyday needs, from parks to grooming and supplies.



Presented To

PADMAPRIYA ARASANIPALAI KANDHADAI

Presented By

CHARLIE MEDIALDEA

Table of Contents

Introduction

I. Introducing the domain, and overview of the background, and context of the research

FurBot was inspired by a personal journey as a dog owner migrating to Canada and facing challenges in navigating an unfamiliar environment. Without local connections, finding dog-friendly facilities, reliable grooming services, and essential pet care resources was overwhelming. This experience revealed a significant gap in accessible, localized, and reliable information for pet owners, particularly in urban settings. It became evident that many dog owners, especially newcomers, share similar struggles in identifying pet-friendly options and resources.

Combining pet care and technology, FurBot is a web-based chatbot designed to provide pet owners with quick and reliable access to essential services and information. FurBot uses a straightforward approach by leveraging predefined intents and responses to handle user queries. Through natural language processing (NLP), the chatbot is trained to recognize user input and match it to the most relevant predefined answers, such as locating nearby veterinary clinics, grooming salons, or dog parks. Additionally, FurBot can assist users with process-related questions, such as guidelines for traveling with pets to the US. This approach ensures an efficient and user-friendly experience while consolidating resources into a single platform, helping pet owners save time and navigate pet care services with ease.

II. Problem Statement

Here are some questions that help frame the problem:

- How can a chatbot provide reliable and real-time assistance to dog owners in finding localized pet care resources?
- What solutions can effectively bridge the gap between dog owners and small pet-related businesses?

Dog owners often struggle to access reliable and localized information about pet-friendly facilities and services, such as veterinary clinics, grooming salons, and dog parks. This challenge is further compounded when navigating process-related tasks, like traveling with pets or understanding local pet care regulations. A centralized, chatbot-driven platform like FurBot can address these issues by leveraging predefined intents and responses, supported by natural language processing (NLP).

III. Relevant Literature and Research

During the pandemic, much of the focus shifted toward essential human healthcare services, such as online consultations, often overlooking the need for accessible and reliable pet care solutions. This gap in localized pet care services highlighted the challenges pet owners face in finding nearby veterinary clinics, grooming salons, and pet supply stores. FurBot addresses this gap by consolidating vital pet care information into a single, chatbot-driven platform. By leveraging predefined intents and natural language processing (NLP), FurBot provides a streamlined experience tailored to users' locations, simplifying access to reliable resources and helping pet owners save time while making informed decisions about their pets' needs.

Advancements in chatbot technology have demonstrated the potential to improve accessibility and convenience for pet owners. Platforms like Woof.AI offer personalized product recommendations to enhance user experiences (Woof.AI, 2024). Similarly, AI-powered veterinary chatbots support scheduling, answer frequently asked questions, and manage reminders for vaccinations, providing round-the-clock assistance (FastBots, 2024). Building on these innovations, FurBot offers a user-friendly and efficient platform that provides real-time assistance through predefined questions and answers, supported by NLP. It ensures users receive accurate, personalized, and location-specific recommendations, meeting a wide range of needs, including finding pet-friendly spaces, accessing essential services, and addressing process-related queries like traveling with pets.

Pet-friendly Parks

Pet Stores

Pet Salon

Pet Hospital

IV. Hypothesis, Assumptions, and Potential Benefit of Research

Hypotheses and Assumptions

FurBot aims to address key challenges pet owners face in accessing reliable, localized pet care services. Localized recommendations will reduce search time for services like veterinary clinics, grooming salons, and dog parks, while personalized suggestions tailored to user profiles will enhance satisfaction. The interactive Q&A chatbot will improve engagement by providing real-time answers to common pet care questions, ensuring a seamless user experience.
The development of FurBot assumes that pet owners prefer a centralized platform offering quick access to multiple pet-related services, as well as personalized and location-specific recommendations. It also assumes that real-time assistance via a chatbot is more practical and engaging than traditional search methods.

Expected Benefits

For users, FurBot simplifies pet care by acting as a one-stop platform, saving time and offering tailored recommendations. Real-time assistance enhances usability, making FurBot a dependable tool for pet owners. For the community, FurBot increases visibility for local pet service providers and improves access to timely resources, fostering better pet welfare and stronger connections within local communities.

Proposed Research Project

I. Research Design and Objectives

Design

Develop a web-based chatbot that consolidates pet-related services into a single platform, offering localized recommendations, personalized suggestions, and a predefined interactive Q&A system. By leveraging natural language processing (NLP) and a dataset of frequently asked questions, FurBot aims to improve accessibility, user satisfaction, and engagement.

Objectives

1. Database Creation: Develop a comprehensive database to provide localized recommendations for pet-related services, including veterinary clinics, grooming salons, and dog parks.
2. User Profile Integration: Implement a user profile system that enables personalized recommendations based on pet owners' preferences and needs, aligned with their location and service requirements.
3. Interactive Q&A System: Design a predefined question-and-answer system that delivers real-time assistance for common pet care inquiries, supported by structured datasets to ensure accuracy and relevance.

II. Methodology

Data Collection

- Gather data from publicly available directories, including information on veterinary clinics, pet grooming salons, and dog parks.
- Research blogs, forums, and articles detailing personal experiences of pet owners, such as traveling with pets, finding pet-friendly accommodations, or accessing pet care services in new locations (e.g., "Bringing your pet to the US for a quick vacation").

Data Analysis

Analyze user behavior and preferences to refine recommendations. Predefined Q&A content will be structured based on the most frequently asked questions by pet owners.

Justification

A consolidated platform provides efficiency and convenience, addressing a common gap in existing tools. The use of a chatbot ensures real-time interaction and tailored assistance, distinguishing FurBot from traditional search methods.

III. Technologies

Operating System

- Windows 11: The development environment is based on Windows 11.

Programming Languages/Frameworks

- Backend: Python will be used for the chatbot’s core functionality, particularly for NLP-based intent recognition and entity extraction. If time permits, OpenAI's pre-trained models will be integrated to enhance conversational capabilities and overall user experience.
- Frontend: The user interface will be developed using HTML, CSS, and JavaScript. Flask will serve as the web framework to connect the backend with the frontend, ensuring seamless integration and functionality.

Database

- SQLite: A lightweight and efficient database will be used to store essential data such as user profiles, localized pet service information, and chatbot interactions.

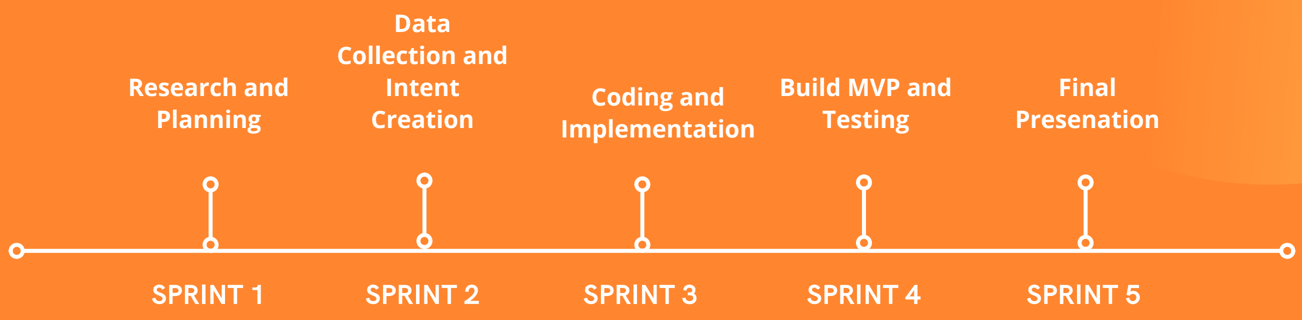
Development Tools and Methodologies

- Version Control with GitHub: GitHub will be used for version control and repository management.
- Agile Methodologies with JIRA: JIRA will be utilized for sprint planning, tracking progress, and managing tasks to ensure iterative development and timely delivery of project milestones.

IV. Expected Results

- Accurate and personalized recommendations for pet-related services, such as veterinary clinics, grooming salons, and dog parks, tailored to user profiles and preferences.
- Practical applications, including a user-friendly chatbot that provides quick, reliable answers to frequently asked pet care questions, making it a valuable tool for pet owners.
- Enhanced user experience through the consolidation of services and real-time assistance, saving time and effort for users while improving access to essential pet care resources.

Project Planning and Timeline



Sprint	Weeks	Milestone
Sprint 1	1 week	Complete research and planning, define project scope, objectives, and requirements. Set up project tools (e.g., GitHub, JIRA).
Sprint 2	1 week	Collect data for pet-related services, define chatbot intents.
Sprint 3	4 weeks	Begin coding and implementation: develop the backend (NLP-based chatbot functionality, SQLite database) and frontend (HTML/CSS/JavaScript with Flask integration).
Sprint 4	4 weeks	Build the minimum viable product (MVP), conduct functional testing, and refine features based on test results.
Sprint 5	1 week	Finalize the app, address remaining issues, and prepare for the official app launch.

Work Date/Hours Logs

Date Worked On	Issue key	Summary	Hours Work	Status
18-01-2025	SCRUM-13	Research a topic for the project proposal	2	Done
18-01-2025	SCRUM-7	Research and Identify Technologies for FurBot	2	Done
19-01-2025	SCRUM-6	Set Up Project Repository	1	Done
19-01-2025	SCRUM-2	Set Up JIRA for Project Management	2	Done
25-01-2025	SCRUM-1	Draft Project Proposal	6	Done

Project Contract

DOUGLASCOLLEGE

January 26, 2025

Padmapriya Arasanipalai Kandhadai
Instructor/Advisor
Douglas College
700 Royal Avenue
New Westminster BC V3M 5Z5

Dear Priya,

This document confirms the commitment to the development and delivery of FurBot: A Web-Based Chatbot for Pet Care Assistance. As described in the proposal, the project aims to create a user-friendly chatbot that integrates personalized recommendations, an interactive Q&A system, and localized service suggestions into a single cohesive platform.

The project will adhere to the timeline outlined in the proposal, with key milestones including the completion of a functional chatbot prototype, enhancement of features, and the final deployment of the application. I will be solely responsible for all phases of development, testing, and deployment, ensuring that the agreed-upon scope and deadlines are consistently met. To maintain steady progress, I will allocate a minimum of 10–12 hours per week to the project during its critical phases.

Regular weekly check-ins will be scheduled to track progress and ensure alignment with the project plan. Any necessary adjustments to the timeline will be communicated in a timely manner. This document reflects my dedication to delivering an innovative and impactful solution that simplifies the way users access essential pet care resources.

Sincerely,

Charlie Medialdea
300374504
Student, Founder & CEO

References

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- Veterinary Chatbots - FastBots. (2024). Enhancing pet care with AI: The role of veterinary chatbots. Retrieved from https://www.fastbots.ai/blog

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- CodeWithHarry. (n.d.). Python for beginners - Learn Python in 1 hour [Video]. YouTube. https://www.youtube.com/watch?v=RpWeNzfSUHw&list=PLqnslRFeH2UrFW4AUgn-eY37qOAWQpJyg

Appendix

- **GitHub Repo** - https://github.com/charlz1202/W25_4495_S1_CharlieM
- **JIRA Project** - https://charliemedialdea.atlassian.net/jira/software/projects/SCRUM/summary